

Chart rules: lines, aspect ratios, heatmaps, type

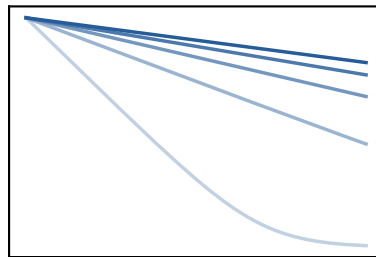
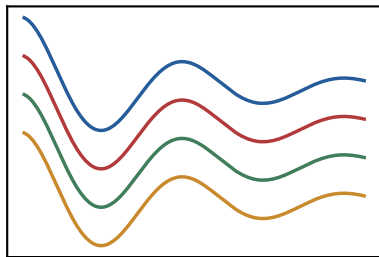
line colors, in order

	blue:	#255C99	1st series: the result the panel is about
	red:	#B33A3A	2nd series: the contrasting case
	green:	#3F7D5B	3rd series
	amber:	#C98A2D	4th series
	purple:	#7B5EA7	5th series (rare: consider panels instead)
	grey:	#8A949C	context and reference curves
	black:	#000000	analytic ground truth only

solid = model or result dashed = derived / reference
dotted = guides and thresholds markers = measured data

categories: palette order

ordered series: one-hue ramp



type in figures

sans-serif in figures; the body text is serif – annotation reads as annotation

author at print width: figsize width = printed width, so pt in the script = pt on the page

oversize canvases: scale fonts with `style_for()`; never shrink a finished figure in LaTeX

10 pt

panel title

9.5 pt

axis labels

8.5 pt

tick labels

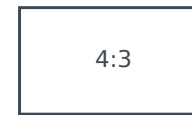
8.2 pt

legend, annotations

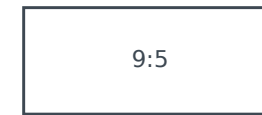
7 pt

floor – nothing smaller

aspect ratios: a menu, not a choice



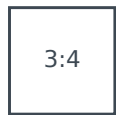
default single
x–y chart



long scans,
spectra, kinetics



maps, parity,
phasors



depth profiles
(z downward)

multi-panel: combine menu items (one ratio per figure; strips excepted)



two 4:3
compare cases



9:5 + strip
fit + residuals



three 1:1
map series

1:1 + strip
map + cut

heatmaps: one map per meaning



viridis: positive scalar fields: intensity, $|E|^2$, visibility
perceptually uniform, colorblind-safe, survives grayscale



inferno: log-scaled detector counts
black floor reads as zero counts



RdBu_r: signed fields: residuals, strain, Δ from reference
diverging; ALWAYS symmetric limits so white means zero



twilight: phase, azimuth, orientation
cyclic: the two ends meet



gray: radiographs and topographs
domain convention; reads as image, not field

label every colorbar with units; log scales get decade ticks
never jet/rainbow (luminance reversals invent edges)

reference: matplotlib.org/stable/users/explain/colors/colormaps.html